SOLDERING TOOLS AND TECHNIQUES

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Abstract

In this SOP, the principle of operation of electrical soldering and the comparison of Surface Mount Technology and Through Hole Technology are discussed. The available soldering equipment at the lab and working procedure are documented.

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1 Electrical Soldering Principle of Operation

1.1 Definition of Electrical Soldering

Electrical soldering is a technique that connects two or more electrical components using solder; it is commonly used to connect electronic components to a circuit board. The solder is made of tin, lead, cadmium, zinc and indium, forming strong electrical bonds between the electronic components. [1, 2]

1.2 Surface Mount Technology (SMT)

Surface mount technology was created due to the high costs and difficulties with fitting the electronic components' leads through the holes on the circuit board. Since it is not necessary for the components' leads to pass through the circuit board holes, surface mount technology is widely used for printed circuit board (PCB) assembly. It allows the SMT components to be designed smaller, which reduces the cost of materials and time. [3]

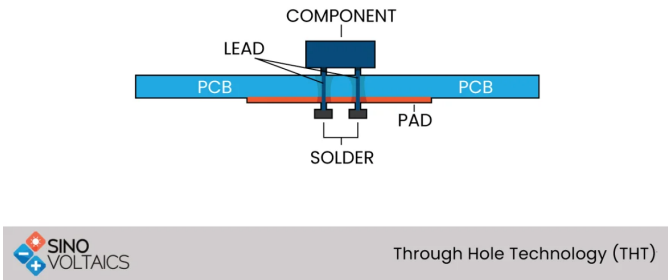

Figure 1: Surface mount technology component mounted directly on a PCB.

The advantages of surface mount technology include low spurious capacitance and inductance, lower power ratings, and creating smaller or more dense circuits. These advantages allow the technologies to adapt quickly with more capabilities.

1.3 Through Hole Technology (THT)

Through hole technology refers to connecting pin-through-hole electronic components to printed circuit boards by mounting the leads of the components to the opposite side of the holes. [4]

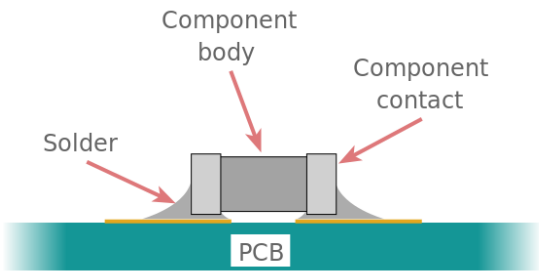

Figure 2: Through hole component leads passed through a PCB and soldered to pads.

The advantages of using through hole technology are that the strong mechanical bonds created by inserting the leads into the holes provide high reliability, and its compatibility

with common packages such as dual-inline package (DIP), plastic DIP (PDIP), zig-zag in-line package (ZIP), or transistor outline (TO) package provides better functionality for the circuit board. [4, 5]

1.4 SMT vs. THT

Table 1: SMT vs. THT [6]

SMT	Versus	THT
Low power	Board Application	High power
May fail in high stress	Environment	Good for high stress
Less expensive at scale	Manufacturing Cost	More expensive in general
Prototyping or production	Development Level	Proof of concept or prototyping
Double sided, small	Board Layout Area	Single-sided, large

Based on the characteristics of surface mount technology and through hole technology, decisions regarding which technology to use could be made. For example, through hole technology will be ideal for military grade PCB or critical system design that has physical and thermal requirements, and surface mount technology will be ideal for tiny consumer electronics devices with a limited cost. [6]

2 Electrical Soldering Tools

2.1 Types of Solder

The three main types of solders are lead based solder, lead free solder, and flux core solder. The different types and ratios of metals that make up the solder result in different melting points and bonding strength.

Table 2: Comparison between solders [7, 8, 9]

Lead Based Solder	Lead Free Solder	Flux Core Solder
Base components: lead + tin **Melting point:** 180–190°C • Shock resistant • Flows well • Lead eliminates tin whiskers • Creates environmental and health hazards	**Base components:** copper + tin + silver + nickel + zinc + bismuth + antimony **Melting point:** 217°C • Requires proper board layout and component selection to prevent damage to electronic components • Better for the environment and human body compared to lead based solder • Cost increases as the concentration of tin increases	**Base components:** metals + flux, typically rosin **Melting point:** 183°C • May contain lead depending on the product itself • Allows for cleaner electrical connections • Flux reduces metal oxidation and promotes the formation of electrical bonds

2.2 Types of Soldering Iron

The different types of soldering irons help to perform different tasks with varied size and temperature. The common types of soldering iron are regulated soldering stations, simple

regulated soldering irons, and simple unregulated soldering irons. The regulated soldering stations offer a large variety of interchangeable tips and a wide range of temperatures. The regulated soldering irons have adjustable knobs to adjust temperatures. However, due to their lack of flexibility with the shape of the tips, they are not ideal for complex tasks. The unregulated soldering irons are cheap and effective. However, due to their lack of flexibility with changing tips and temperatures, they are not widely used for beginners or to complete complex tasks. [10]

3 Electrical Soldering in CaYPT Lab

The CaYPT lab operates Hakko FX-951 along with SainSmart Portable Fume Extractor and 60Sn/40Pb lead solder.

- User Manual: https://www.hakko.com/english/support/doc/result.php?s_cat[1]=1&s_keyword=FX-951#resultAnc
- Soldering Station Order Link: https://www.hakko.com/english/products/hakko_fx951.html
- Smoke Absorber Order Link: https://www.amazon.ca/SainSmart-Magnetic-Solder-Smoke-Absorber/dp/B0C8N64N5N/ref=sr_1_9?crid=2UDZOF62TGXCJ&dib=eyJ2Ijoi MSJ9.Vcf_e03BIJ1dUWCRgpIXHZz_ZMLpV8qxZ1G65U_VbJnakiMWxTTONo6QiyCqnXZ9An4kEOBW_S2rN10YrGUjGzQqQdCaRum8iro7Ar-rSBL2wApJLNNVwtvOlMKqftxSsZ6E4dDW2T_3iBImToLKp7-IBNFC_CA7sJ-Z6oySJ0XTPCtW4YKtPNKwB6XTxI_FEgaUwertSJyNawAq0sTEPbSAhTgcc_1XitVwshRks5n2kbmphzEasG3RYH87xuB3rWr4W57684bLvjEFgaq54PSfSO47xnklCEqknOpocW4.1I2AfIY0DAm93raaKbocaS0P_WeVF1-qD35hto z1iNc&dib_tag=se&keywords=soldering+fan&qid=1777920031&sprefix=soldering+fan%2Caps%2C113&sr=8-9

4 General Procedure of Operation

4.1 Soldering Station

1. Prepare all the materials needed.
2. Select a suitable tip for the soldering project.
3. Heat up the soldering station to a selected temperature.
4. Tin the soldering iron tip with solder.
5. Melt the solder and connect the electrical components.

4.2 Precautions

1. Wear proper PPEs.
2. Do not use solder with acid unless specifically directed.
3. Place the soldering iron back on the station stand after using it.
4. Hold small electronic pieces with clamps or tweezers.

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